

The logo consists of three vertical bars of increasing height from left to right. The first bar is yellow, the second is red, and the third is blue.

ProBUILD

Week 2



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Assignment_2.1

FreeCAD 1.0.2 - Unnamed

File Edit View Tools Macro Sketch Windows Help

Sketcher

Unnamed: 1*

Spreadsheet

Content: 'centre_cir_dia' Alias:

	A	B	C	D	E
7	bottom_right_arc_radi	22			
8	bottom_cir_dia	-			
9	bottom_left_lower_arc_radi	16			
10	bottom_left_middle_arc_radi	35			
11	bottom_left_upper_arc_radi	16			
12	middle_left_outer_cir_dia	-			
13	top_left_arc_radi	44			
14	slot_radi	11			
15	slot_angle	40.00 deg			
16	bottom_inner_cir_dia	-			
17	middle_left_inner_cir_dia				
18	centre_to_bottom_cir_ydist				
19	centre_to_top_cir_ydist				
20	left_inner_cir_to_top_inner_cir_ydis				
21	left inner cir to top inner cir xdis				

Model

Unnamed

Body

Origin

Sketch

Spreadsheet

Tasks

Close

Solver messages

Fully constrained

Constraints

Filters

Constraint1

Constraint2

Constraint3

Report view

17:32:10 <Sketch> SketchObject.cpp(11403): invalid subname ;g2v1;SKT.vertex2
 17:32:11 <Sketch> SketchObject.cpp(11403): invalid subname ;g2v1;SKT.
 17:32:11 <Sketch> SketchObject.cpp(11403): invalid subname ;g2v1;SKT.

Preselected: Unnamed.Body.Sketch.H_Axis (-126.88 mm, 0.00 mm, 2.00 µm)



Assignment_2.2

FreeCAD 1.0.2 interface showing a sketch of a mechanical part and a spreadsheet with dimensions.

Sketch: A 2D sketch of a mechanical part, likely a bracket or a connector, featuring a central circular hole and four smaller circular holes arranged symmetrically. The sketch is centered on a coordinate system with X, Y, and Z axes.

Spreadsheet: A table containing dimensions and constraints for the sketch.

	A	B	C	D	E	F
1	c1_dia	30	R15	6x		
2	c2_dia	42	R21	5x		
3	c3_dia	40	R20	2x		
4	centre_cir_dia_1	32	R16			
5	centre_cir_3	64	R32			
6	centre_cir_4	80	R40			
7	centre_to_top_cir_ydist	60				
8	centre_to_top_cir_xdist	50				
9	centre_to_middle_cir_xdist	70				
10	curv_radi	20				

Model: A tree view showing the hierarchy of the model. The 'Body' is the root, followed by 'Origin', 'Sketch', and 'Spreadsheet'.

Tasks: A panel showing the status of the sketch. It indicates 'Fully constrained' and lists constraints: Constraint1, Constraint2, and Constraint3.

Report view: A panel showing the status of the sketch. It displays the following messages:

```

18:22:34 <Sketch> SketchObject.cpp(11403): invalid subname ;g5v1;SKT.Vertex3
18:22:34 <Sketch> SketchObject.cpp(11403): invalid subname ;g5v1;SKT.Vertex3
18:22:34 <Sketch> SketchObject.cpp(11403): invalid subname ;g5v1;SKT.
  
```

Preselected: Unnamed.Body.Sketch.Edge6 (58.04 mm, 46.66 mm, 8.00 µm)



Learnings

- 1) How Identify Missing Constraints*
- 2) Linking name and dimension (aliases)*
- 3) Linking spreadsheet to sketch*

Difficulties

- 1) Alias naming confusion*
- 2) Difficult to track which value controls which part*
- 3) Time consuming At beginning*